1. Respiration

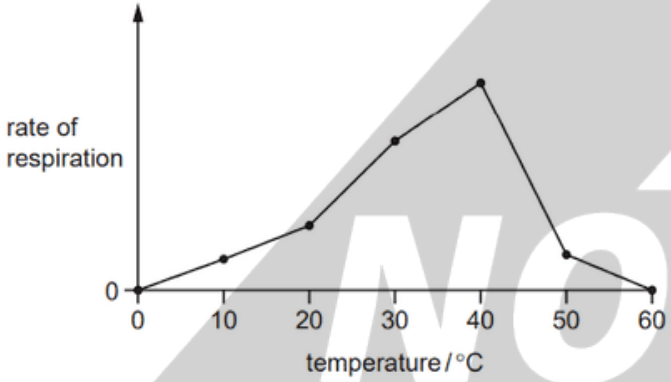
1.1. Respiration

Respiration: Chemical reactions that break down nutrient molecules in living cells to release energy.

- Uses of energy in the body of humans: muscle contraction, protein synthesis, cell division, active transport, growth, the passage of nerve impulses and the maintenance of a constant body temperature.
- Respiration involves the action of **enzymes** in cells to speed up the reaction.

Effect of Temperature on Respiration in Yeast

- An indicator, **Methylene Blue**, can be used to investigate the effect of temperature on yeast cells.
- Yeast can respire both aerobically and anaerobically, though, in this experiment, their aerobic respiration rate is being investigated.
- The time taken for the methylene blue to lose its colour measures the respiration rate of the yeast cells in the suspension.
  - The faster the dye changes from blue to colourless, the faster the respiration rate.
- IV: Effect of Temperature
- DV: Rate of Respiration (recording the **time taken** for methylene blue dye to change from **blue to colourless)**



Exam Tip! In the exam, always state that energy is **released**; it is NEVER made, produced, or created.

1.2. Aerobic Respiration

Aerobic Respiration: chemical reactions in cells that use oxygen to break down nutrient molecules to release energy

$$Glucose + Oxygen \rightarrow CarbonDioxide + Water$$

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$$

1.3. Anaerobic Respiration

Anaerobic Respiration: chemical reactions in cells break down nutrient molecules to release energy without using oxygen.

- **In muscles** (vigorous exercise):
  - $Glucose \rightarrow Lactic\ Acid$
- **In yeast** (single-cell fungi)**:**
  - $Glucose \rightarrow Ethanol + Carbon\ Dioxide$
  - $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$
- **Disadvantages of Anaerobic Respiration:**
  - Only produces 1/20 of the energy per glucose molecule that aerobic respiration would
  - Produces poisonous lactic acid
- **Lactic Acid:**
  - Builds up in muscles and blood during vigorous exercise
  - The heart, liver and kidneys need extra oxygen to do this, which causes you to continue breathing heavily after exercise.
  - The extra oxygen is called the oxygen debt.
- **Oxygen Debt is removed by:**
  - continuation of fast heart rate to transport lactic acid in the blood from the muscles to the liver
  - continuation of deeper and faster breathing to supply oxygen for aerobic respiration of lactic acid
  - aerobic respiration of lactic acid in the liver

1.4. Comparison of Aerobic and Anaerobic Respiration

	Aerobic	Anaerobic
Oxygen	Needed	Not needed
Breakdown of Glucose	Complete combustion	Incomplete combustion
Products	Carbon Dioxide and Water	**Animals**: Lactic Acid & **Yeast**: Carbon Dioxide and Ethanol
Amount of Energy Released	More	Less

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